Section 1 : Choose the correct answer (6 points)

1-) Random Access Memory is volatile storage because:

A) It loses all stored data instantly when electrical power is disconnected
B) Its speed changes unpredictably depending on software load
C) Data can only be written once and never modified
D) It is highly fragile and physically breaks easily

2-) What is the primary role of the Central Processing Unit:

A) To permanently save files when the machine loses power
B) To connect the computer to local networks and routers
C) To display monitor output
D) To execute sequences of instructions

3-) Python is interpreted because:

A) A compiler translates the code into instructions the CPU can run
B) A program called the interpreter executes the code line by line
C) It bypasses the CPU entirely and executes code directly inside the computer's RAM

Section 2 (3 points):

1-) Convert 01000011 to base-10
2-) Compare arguments and parameters in a function
3-) Compare the integer and a float data-type
4-) Copy and complete the truth table

a	b	a **AND** b	a **OR** b	**NOT** a	**NOT** b
False	False				
True	False				
True	True				
False	True				

Section 3: Identify the pieces of code (3 points)

For each of the functions below, determine/identify the following:

- The name of the function
- The parameters they accept
- The procedure (steps) the function does
- The output (result) of the procedure

Function A:

```
def divide(a,b):
    return a / b
```

Function B:

```
def greet(name):
    greeting = "Good morning " + name
    return greeting
```

Function C:

```
def subtract_two(number):
    return number - 2
```

Function D:

```
def give_last_element(num_list, list_length):
    return num_list[list_length - 1]
```

Function E:

```
def does_name_exist(names_list, target_name):
    does_exist = False

    for name in names_list:
        if name == target_name:
            does_exist = True

    return does_exist
```

Function F:

```
def is_adult(age):
    if age >= 18:
        return True
    else:
        return False
```

Section 4: From The Text (8 points)

General Instruction Types

Instructions are grouped into ordered lists that, when taken as a whole, tell the different parts of the computer how to work together to perform a specific task, like grayscaling an image or playing a media file. These ordered lists of instructions are called *programs*, and they consist of a few basic types of instructions.

In modern RISC microprocessors, the act of moving data between memory and the registers is under the explicit control of the code stream, or program. So if a programmer wants to add two numbers that are located in main memory and then store the result back in main memory, he or she must write a list of instructions (a program) to tell the computer exactly what to do. The program must consist of:

- a `load` instruction to move the two numbers from memory into the registers
- an `add` instruction to tell the ALU to add the two numbers
- a `store` instruction to tell the computer to place the result of the addition back into memory, overwriting whatever was previously there

These operations fall into two main categories:

Arithmetic instructions
These instructions tell the ALU to perform an arithmetic calculation (for example, `add`, `sub`, `mul`, `div`).

Memory-access instructions
These instructions tell the parts of the processor that deal with main memory to move data from and to main memory (for example, `load` and `store`).

Figure 1: Inside The Machine - Jon Stokes

1-) State 4 examples of arithmetic instructions?
2-) State 2 memory-access instructions?
3-) What does a load, add, and store instruction tell the computer to do?
4-) What does the text refer to an ordered list of instructions as?

Section 5: Bonus

1-) What is a deterministic function? State an example of a function that is deterministic, and why?
2-) Write 4 examples of comparison operators
3-) The following function takes an n-length number list, express the total amount of arithmetic operations done by the function(algorithm) in terms of n
(Note: Each addition, subtraction, division and multiplication is an arithmetic operation and the only operation that should be accounted for)

```
def calculate_each_element(num_list, list_length):
    for number in num_list:
        first_calculation = number + list_length
        second_calculation = first_calculation - number
        third_calculation = second_calculation / 2
        print(third_calculation)
    return "Finished calculating"
```